

```
print('hello world')
```

hello world

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
df = pd.read_csv("train.csv")
```

```
df.head()
df.shape
df.info()
df.describe()
df.isnull().sum()
df.nunique()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 12 columns):
```

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

```
dtypes: float64(2), int64(5), object(5)
```

```
memory usage: 83.7+ KB
```

PassengerId	891
Survived	2
Pclass	3
Name	891
Sex	2
Age	88
SibSp	7
Parch	7
Ticket	681
Fare	248
Cabin	147

```

Embarked      3
dtype: int64

df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

# Fix chained assignment warnings
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

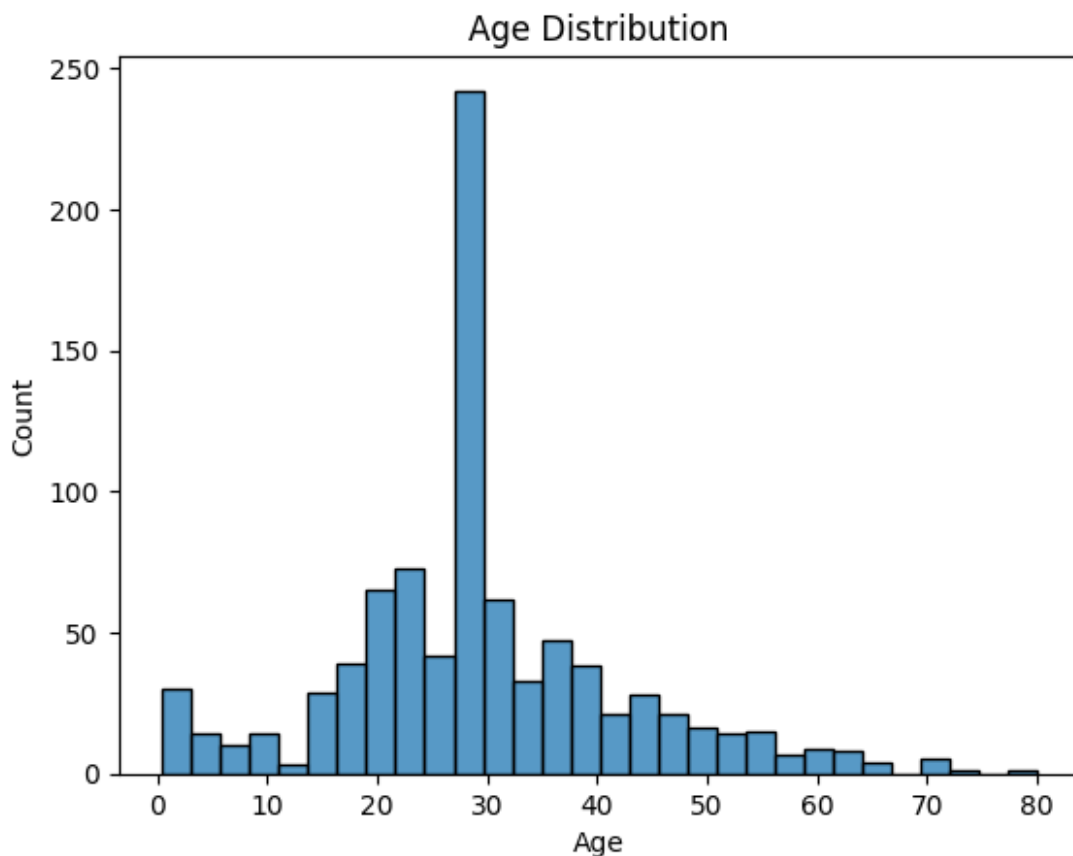
# Only drop 'Cabin' if it exists
if 'Cabin' in df.columns:
    df.drop(columns=['Cabin'], inplace=True)

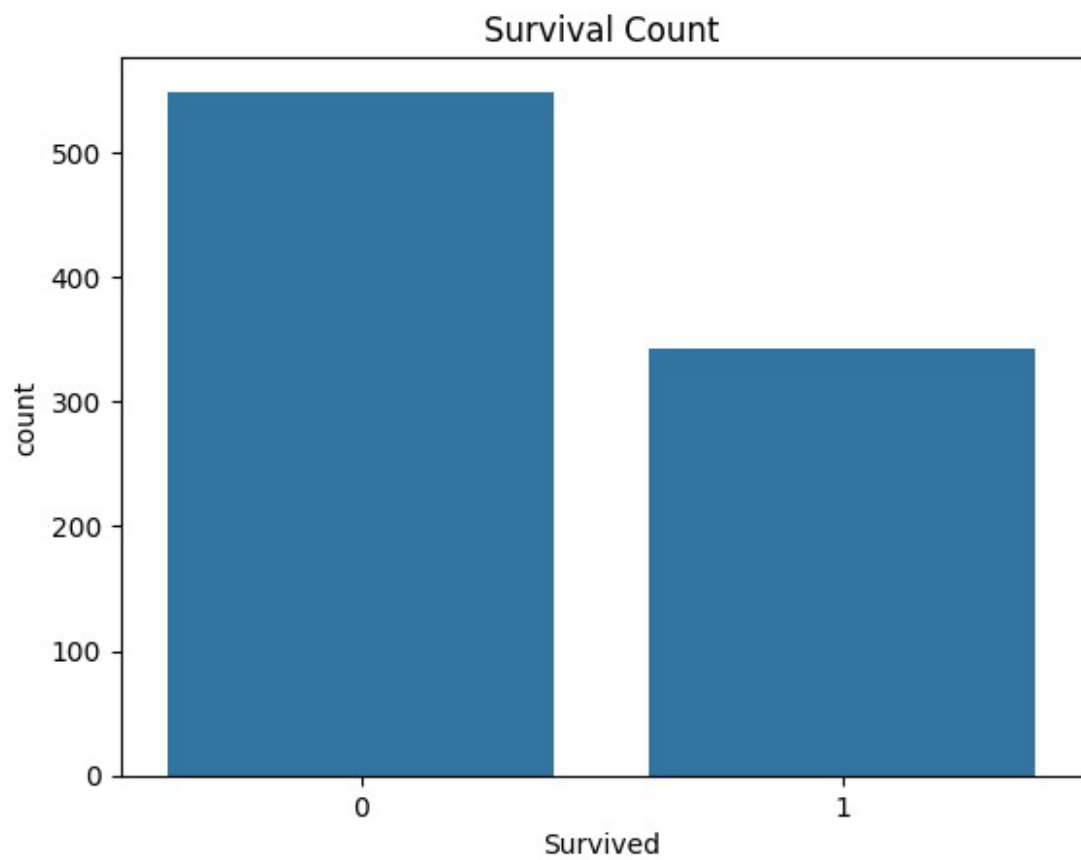
sns.histplot(df['Age'], bins=30)
plt.title("Age Distribution")
plt.show()

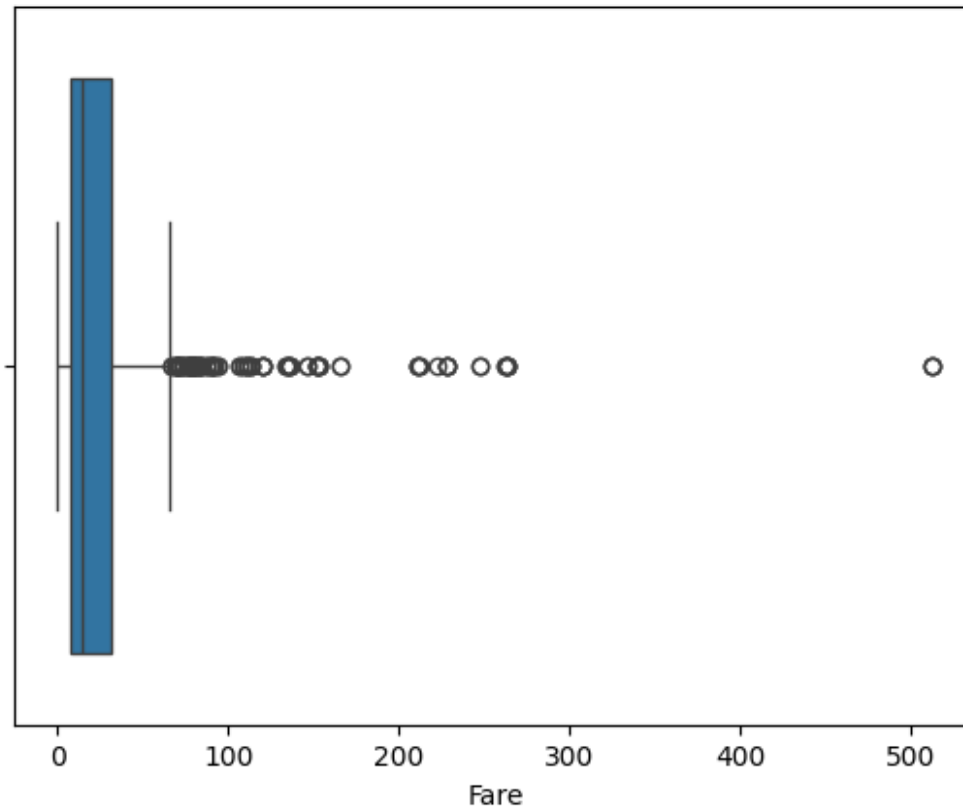
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()

sns.boxplot(x=df['Fare'])
plt.show()

```







```
# Survival by Sex
```

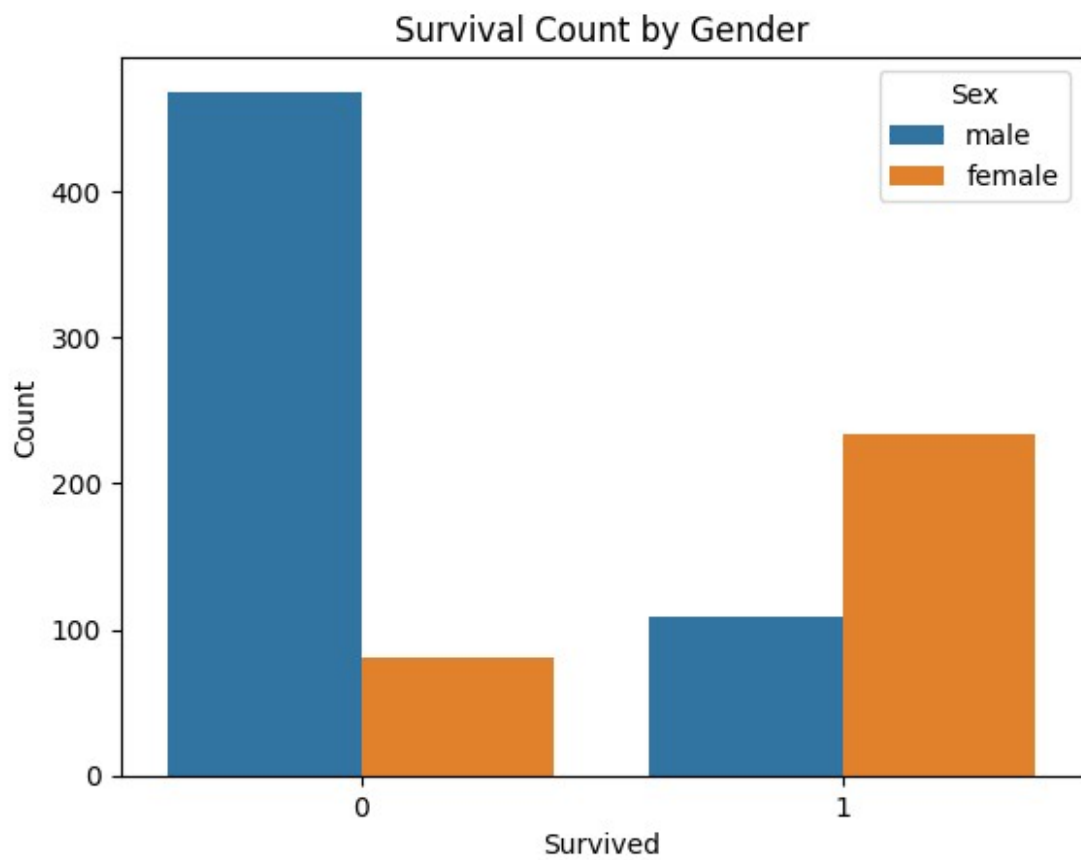
```
sns.countplot(x='Survived', hue='Sex', data=df)
plt.title("Survival Count by Gender")
plt.xlabel("Survived")
plt.ylabel("Count")
plt.show()
```

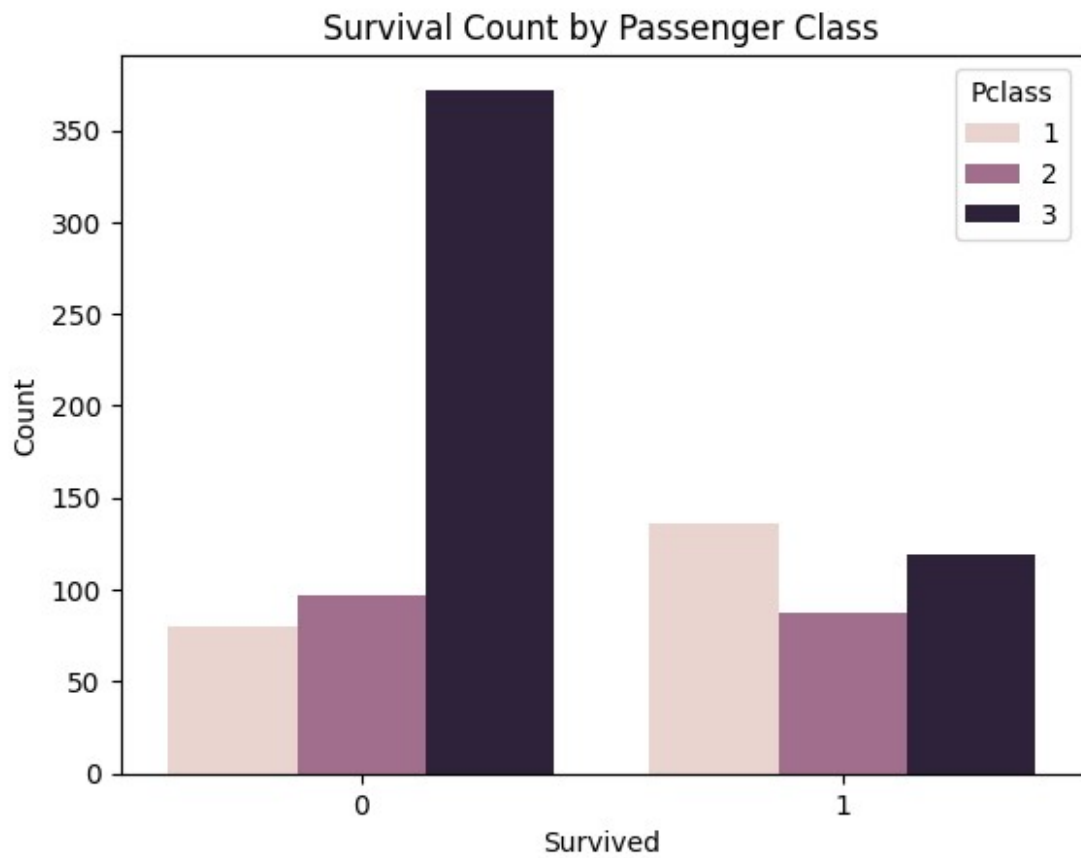
```
# Survival by Pclass
```

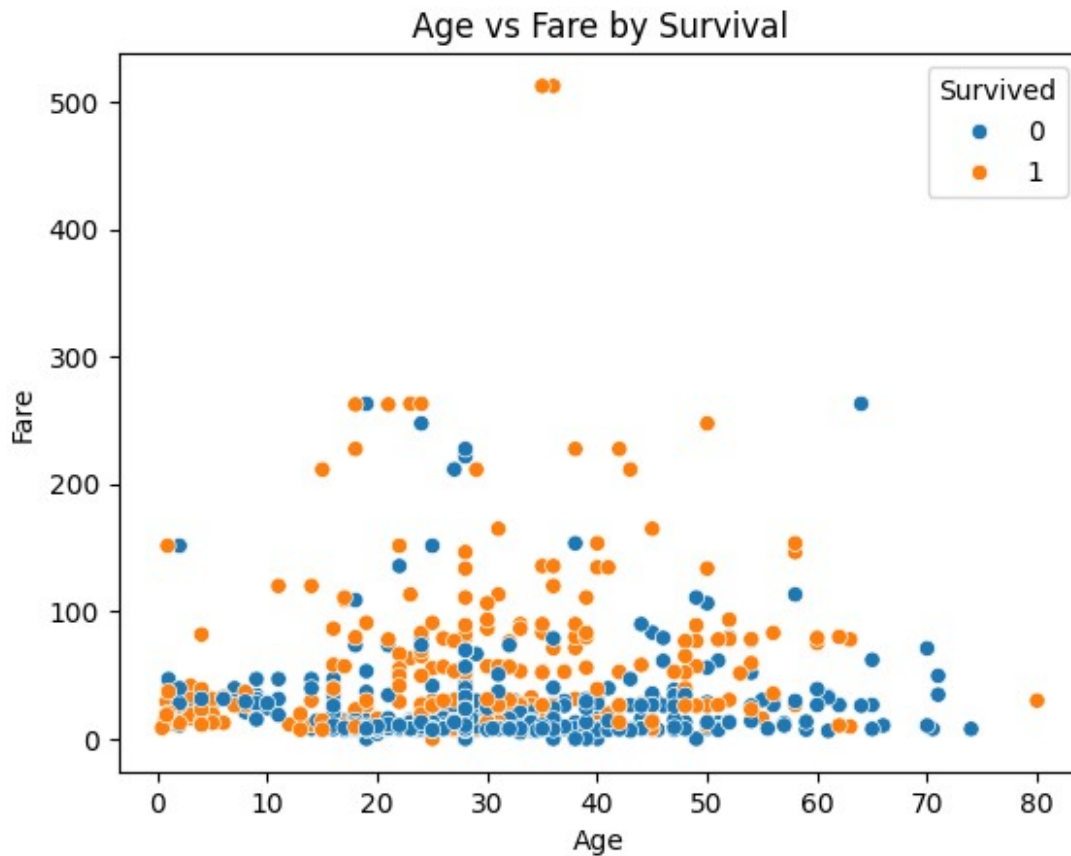
```
sns.countplot(x='Survived', hue='Pclass', data=df)
plt.title("Survival Count by Passenger Class")
plt.xlabel("Survived")
plt.ylabel("Count")
plt.show()
```

```
# Age vs Fare
```

```
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title("Age vs Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```

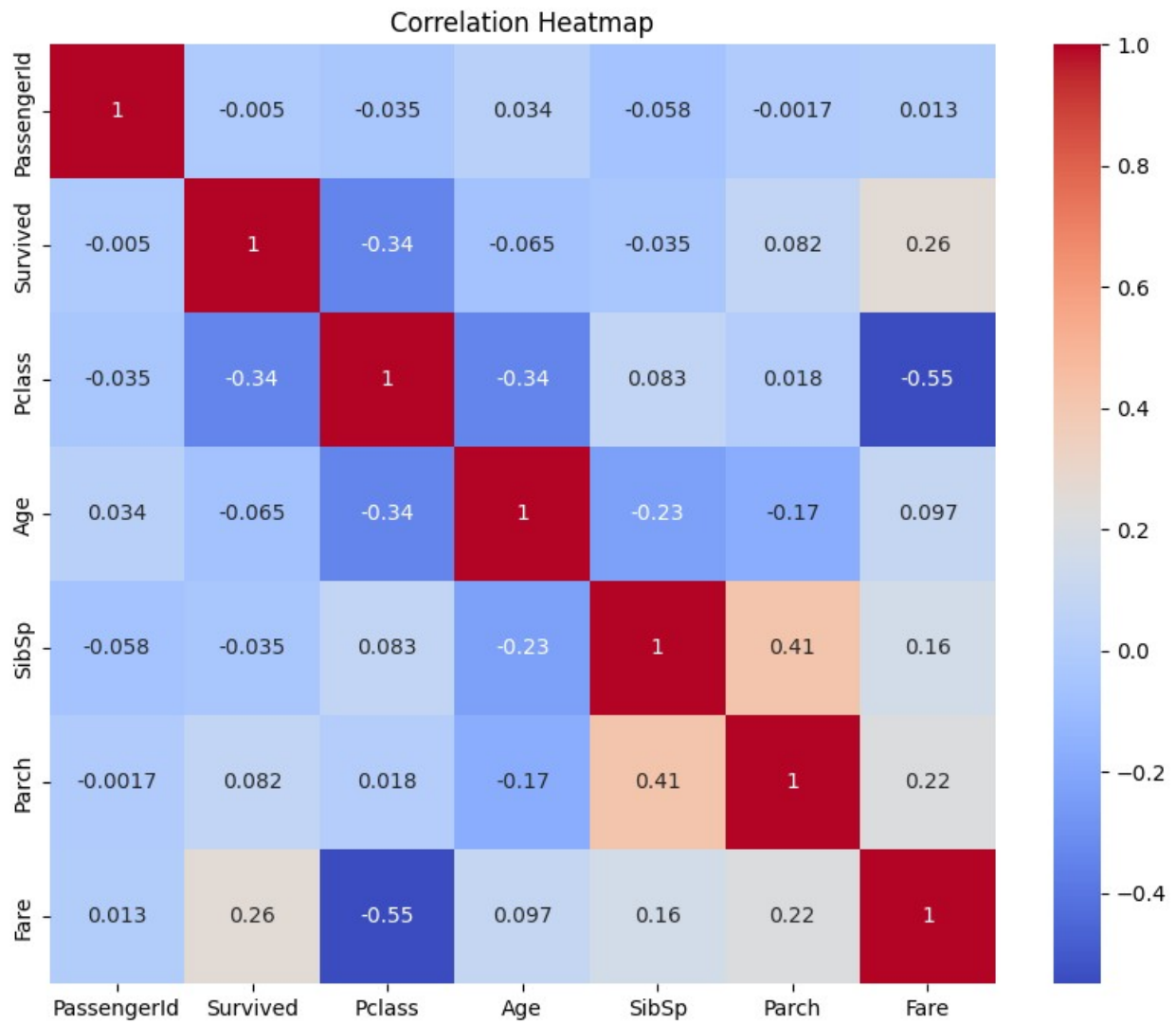






```
# Heatmap of correlations
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()

# Pairplot of selected features
sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']],
hue='Survived')
plt.suptitle("Pairplot: Age, Fare, Pclass vs Survived", y=1.02)
plt.show()
```



Pairplot: Age, Fare, Pclass vs Survived

